

PRE-SOLICITATION NON-COMPETITIVE –

Notice of Intent to Sole Source

1. **SOLICITATION NUMBER: PCA-NIMH-01917**
2. **TITLE:** Muse S3 Athena Headbands and Accessories.
3. **RESPONSE DATE:** April 6, 2026, at 4:00 pm EST.
4. **PRIMARY POINT OF CONTACT:**

Jaddua Johnston

Jaddua.Johnston@nih.gov

Phone: 240-569-0006

I. INTRODUCTION:

THIS IS A PRE-SOLICITATION NON-COMPETITIVE (NOTICE OF INTENT) SYNOPSIS TO AWARD A CONTRACT WITHOUT PROVIDING FOR FULL OR OPEN COMPETITION (INCLUDING BRAND-NAME).

The Genetic Epidemiology Research Branch (GERB) of NIMH is studying the circadian and seasonal rhythms of the internal and external domains and correlates of mood regulation and mood disorders in a community-based sample of families in collaboration with the Experimental Therapeutics Branch of NIMH and the Metabolic Clinical Research Unit of the National Institute of Diabetes and Digestive Diseases (NIDDK) at the NIH Clinical Center. The chief aim is to characterize the inter-relationships of the domains associated with emotional states including sleep, physical activity, diet, cognitive function, and physical health with physiology including heart rate, cortisol, melatonin, and energy metabolism in both clinical and ecological settings. This study is testing the feasibility, reliability and validity of a comprehensive mobile ecological toolkit (MET-c) has been developed for participants that tests these domains in their home environments. This project has already enrolled more than 50 adults and children and now urgently requires additional resources to fulfill the aims of the protocol. This order involves studies of sleep architecture including clinical sleep studies (polysomnography) in the NIH Clinical Center and remote sleep studies in the participant's home environments. After a review of available remote sleep devices, we previously selected and purchased the Muse-S headbands Interaxon Inc., for the MET-c, and have now employed the device in 20 participants with great success. We urgently need to purchase additional units of the upgraded Muse-Athena headbands to maintain study progress.

II. NORTH AMERICAN INDUSTRY CLASSIFICATION SYSTEM (NAICS) CODE and SET ASIDE STATUS

NAICS code 334510- ELECTROMEDICAL AND ELECTROTHERAPEUTIC APPARATUS MANUFACTURING with a business size standard classification of 1000. This acquisition is NOT set aside for small businesses.

III. REGULATORY AUTHORITY

The resultant contract will include all applicable provisions and clauses in effect through the Federal Acquisition Circular (FAC) 2024-07 dated 08-29-2024.

This acquisition is conducted under the procedures as prescribed in FAR subpart 13.5—Simplified Procedures for Certain Commercial Items and FAR subpart 12—Acquisition of Commercial Items at an amount exceeding the SAT, and not exceeding \$7 million.

IV. STATUTORY AUTHORITY

This acquisition is for a commercial item or service and is conducted under the authority of the FAR Subpart 13.5—Simplified Procedures for Certain Commercial Items and the statutory authority of [specify the applicable authority: (1) FAR 6.302-1—Only one responsible source and no other supplies or services will satisfy agency requirements 41 U.S.C. 3304 (a)(1).

V. DESCRIPTION OF REQUIREMENT:

A. PURPOSE AND OBJECTIVES:

The Genetic Epidemiology Research Branch (GERB) in collaboration with the Experimental Therapeutics and Pathophysiology Branch (ETPB) of the NIMH, and the Metabolic Clinical Research Unit of the National Institute of Diabetes and Digestive Diseases (NIDDK) at the NIH Clinical Center is studying the inter-relationships between emotional states and homeostatic rhythms of sleep, physical activity, diet, cognitive function, and physical health and their physiologic and environmental correlates. Most of the research in these areas is based on studies in laboratory settings that often do not patterns observed in real life settings. This work has important implication for both clinical and population level interventions. In particular, sleep studies conducted in classical sleep laboratories do not reflect sleep patterns in daily life. We have now collected substantial amount of data in both settings to begin to address this gap in information about sleep and physical activity that can be used for interventions and prevention. We have applied the Muse-Athena headband, developed by Interaxon Inc., a small, lightweight, wearable, wireless, untethered headband that is uniquely designed to

measure sleep in an outpatient (home) setting in more than a dozen participants. We now have a need to purchase more units to continue the pace of recruitment of patients at the NIH Clinical Center who have enrolled in the MET-c component of the study. Use of these devices will enable us to collect more ecologically valid data and minimize the need to stay overnight at the NIH Clinical Center.

Purchase Description:

Muse-Athena Headbands, Replacement headband straps, data services and subscription (data processing and technical support)

5. Salient characteristics:

Signal Specifications:	
Biofeedback Sensors	Strength neurofeedback training (fNIRS); brain oxygenation tracking (fNIRS); digital sleeping pill (EEG+PPG+Smart-Fade technology); overnight tracking insights (EEG, PPG, accelerometer, gyroscope); mind (EEG); heart (PPG); breath (PPG, accelerometer, gyroscope); body (accelerometer, gyroscope)
4x EEG channels	4 EEG channels + 4 amplified Aux channels
fNIRS Sample Rate	5-optode sensor, 64hz
1x PPG channel	Triple wavelength: IR (850nm), Near-IR (730nm), Red (660nm), 64 Hz sample rate, 20-bit resolution
1x Accelerometer	Three axis @ 52Hz, 16-bit resolution, range +/- 4G; 52 Hz sample rate; 16-bit/sample resolution
4x Aux channel connection	Input range; 1.45mV p- p AC coupled signal, with 1.45V DC offset; Aux L on pin3 of USB (USB D+) Aux R on pin4 of USB (ISB ID)
1x Wireless connection	Bluetooth®5.3
Muse App compatibility	iOS 15 and above, Android 8 and above

Performance Specifications:	
Recording Time	Up to 10 hours
Physical Specifications:	
Power Source	Rechargeable Li-ion
Charging Method requirements	USB wall charger delivering a maximum 5v (5 volts) and 2A (2 amps)
Charge time	3 hours
Size range	43-63cm; fits ages 7+, custom smaller bands available
Weight	41g
Materials	Band (23% rayon, 73% polyester, 4.5% spandex); Pod (polycarbonate plastic with soft touch paint, magnets, glue)
EEG Electrode materials	Conductive silver thread
USB 2.0 Connection	Micro-USB

Muse-Athena unique features include:

- **Validated 5-stage sleep scoring: Cohens kappa of 0.76, accurately classifying Stage 1, Stage 2, Sleep, REM, and Awake**
- Microarchitecture of sleep: detailed analysis of sleep microevents including: sleep spindles, k-complexes, deep sleep density

- Heart rate metrics: monitoring cardiovascular activity
- Sleep Metrics: Raw EEG .EDF files for each session; .CSV files for EEG, PPG, and accelerometer data
- Custom reporting and calculations for sleep features including: sleep stages, time spent in stages, sleep time, sleep efficiency, sleep quality, sleep latency
- Mobile application customization: assessments, patient education, videos and step-by-step instructions. Provides real time feedback on signal quality and device fit to participant to ensure quality data.
- Signal quality: signal quality index for each channel per 30 second epoch helps determine if data were good enough to sleep stage.

6. Quantity: 40

B. PERIOD OF PERFORMANCE: One year warranty for the headbands is included.

VI. CONTRACTING WITHOUT PROVIDING FOR FULL OR OPEN COMPETITION (INCLUDING BRAND-NAME) DETERMINATION

The determination by the Government to award a contract without providing for full and open competition is based upon the market research conducted. Specifically, Interaxon Inc. is the only vendor in the marketplace that can provide the services required by NIMH. The equipment is proprietary to its manufacturer, Interaxon Inc, and this company is the sole provider of maintenance for this equipment.

In accordance with FAR part 10, extensive market research was conducted to reach this determination. Additionally, contacts with knowledgeable individuals both in industry and Government, as well as reviews of available product literature, revealed no other sources. Finally, a review of the GSA Advantage, Dynamic Small Business Search, and previous government acquisitions returned no results that meet all of the requirements. Therefore, only Interaxon Inc is capable of meeting the needs of this requirement.

The intended source is:
Interaxon Inc.
22 St. Clair Avenue East
Suite 200
Toronto ON M4T 2S3
Canada

VII. CLOSING STATEMENT

THIS SYNOPSIS IS NOT A REQUEST FOR COMPETITIVE PROPOSALS. However, interested parties may identify their interest and capability to respond to this notice.

A determination by the Government not to compete this proposed contract based

upon responses to this notice is solely within the discretion of the Government. All responsible sources may submit a capability statement, proposal, or quotation which shall be considered by the agency. The information received will normally be considered solely for the purposes of determining whether to proceed on a non-competitive basis or to conduct a competitive procurement. This notice does not obligate the Government to award a contract or otherwise pay for the information provided in response.

Responses to this notice shall contain sufficient information to establish the interested parties' bona-fide capabilities for fulfilling the requirement and include: unit price, list price, shipping and handling costs, the delivery period after contract award, the prompt payment discount terms, the F.O.B. Point (Destination or Origin), the Dun & Bradstreet Number (DUNS), the Taxpayer Identification Number (TIN), and the certification of business size. All offerors must have an active registration in the System for Award Management (SAM) www.sam.gov.

All responses must be received by the closing date and time of this announcement and must reference the solicitation number, **PCA-NIMH-01917**. Responses must be submitted electronically to Jaddua Johnston, Contract Specialist at Jaddua.Johnston@nih.gov.